SOFTWARE SYSTEM PROPOSAL

Bella Glow Studio — Beauty Salon Management System

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1. Business Understanding (DDD — Domain Exploration)

1.1 Core Business Processes

- Appointment scheduling — matching customers with available staff slots for selected services.
- Service delivery — staff performing beauty treatments (hair, nails, skin, brow shaping).
- Payment collection — accepting cash, card, or bank transfer per appointment.
- Customer relationship management — maintaining client history and preferences.
- Staff management — assigning services to qualified employees and tracking schedules.
- Retail sales — selling hair/skin care products at the front desk.
- Reporting & analytics — reviewing revenue, staff performance, and service popularity.

1.2 Main Services Offered

Service	Duration	Specialist
Haircut	30 min	Hairdresser
Hair Coloring	120–180 min	Hairdresser
Blow-dry	30–45 min	Hairdresser
Manicure	45 min	Nail Technician
Pedicure	60 min	Nail Technician
Facial	60 min	Esthetician
Eyebrow Shaping	20–30 min	Esthetician

1.3 Pain Points Without Software

- Double-booking conflicts due to manual paper appointment book.
- High no-show rate (~15–20%) from lack of automated reminders.
- Inaccurate payment records when mixing cash, card, and bank transfers.
- No performance visibility — owner cannot identify top-performing services or employees.
- Staff communication via WhatsApp is informal and unreliable.
- Customer health/allergy information is not systematically stored, posing safety risks.

1.4 Goals of the System

- Eliminate double-bookings through real-time calendar management.
- Reduce no-shows by 70% via automated SMS/email reminders.
- Provide accurate, real-time payment and billing records.
- Enable data-driven decisions through monthly and weekly reports.
- Allow customers to self-book online, reducing front-desk workload.
- Enforce service-to-staff assignment rules automatically.
- Centralize and secure sensitive client data including allergy notes.

2. Functional Features

2.1 Customer Management

The system shall maintain a full customer profile including name, contact details, appointment history, preferred staff member, and health/allergy notes. Staff with appropriate permissions may view and update client profiles.

2.2 Appointment Scheduling

A real-time calendar shall allow booking of appointments, enforcing staff availability and service-to-staff skill matching. Customers may self-book via an online portal, selecting service, preferred employee, and available time slot. The system shall prevent double-booking at the database level.

2.3 Staff Management

Each employee will have a profile listing their name, role, assigned services, working hours, and schedule. The system enforces that only qualified staff are bookable for specific services (e.g., manicures only bookable with nail technicians).

2.4 Payments & Billing

The system shall record each payment linked to an appointment, including amount, method (cash/card/transfer), date, and status (paid/unpaid/partial). Invoices shall be auto-generated per appointment.

2.5 Notifications

Automated notifications shall be sent to customers 24 hours and 2 hours before their appointment via SMS and/or email. Staff shall receive in-app and email alerts upon new bookings, reschedules, and cancellations.

2.6 Reports

The system shall generate reports on revenue by service type, revenue by employee, appointment volume trends, no-show rates, and top customers. Reports shall be filterable by day, week, and month.

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3. Domain Model (Conceptual)

3.1 Main Entities / Aggregates

Entity / Aggregate	Description
Customer	A person who books services at the salon. Aggregate root for booking history.
Employee	A staff member with specific service skills and a weekly schedule.
Service	A named treatment with fixed duration, price, and required skill.
Appointment	Core aggregate — links Customer, Employee, Service, date/time, and status.
Payment	Financial record linked to an Appointment (amount, method, status).
Notification	An outbound message linked to an Appointment (type, channel, sent time).

3.2 Key Relationships

- A Customer can have many Appointments (one-to-many).
- An Employee can have many Appointments (one-to-many).
- An Appointment is linked to exactly one Service (many-to-one).
- An Appointment has exactly one Payment (one-to-one).
- An Appointment may trigger multiple Notifications (one-to-many).
- An Employee is linked to one or more Services they are qualified to perform (many-to-many).

3.3 Important Business Rules

- An employee cannot have two overlapping appointments.
- A service can only be booked with an employee qualified for that service.
- Appointment status flow: Pending → Confirmed → Completed | Cancelled | No-Show.
- A payment record is created automatically when an appointment is confirmed.
- Notifications must be sent at T-24h and T-2h before each confirmed appointment.
- Cancellations within 2 hours of appointment time are flagged as late cancellations.

3.4 Value Objects

- TimeSlot — an immutable pair of (start_time, end_time) used to define availability.
- Money — a combination of (amount, currency) used in Payment and Service pricing.
- ContactInfo — a reusable object containing (phone, email) for both Customer and Employee.

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4. Database Structure (High-Level)

The following tables form the core relational database schema for the system.

Customers

Fields: id (PK), first_name, last_name, phone, email, date_of_birth, preferred_employee_id (FK), allergy_notes, created_at.

Employees

Fields: id (PK), first_name, last_name, phone, email, role, is_active, hire_date.

Services

Fields: id (PK), name, description, duration_minutes, price, category.

Employee_Services

Fields: employee_id (FK), service_id (FK) — junction table.

Appointments

Fields: id (PK), customer_id (FK), employee_id (FK), service_id (FK), start_datetime, end_datetime, status, notes, created_at.

Payments

Fields: id (PK), appointment_id (FK), amount, currency, method (cash/card/transfer), status (paid/unpaid/partial), paid_at.

Notifications

Fields: id (PK), appointment_id (FK), type (reminder/confirmation/cancellation), channel (sms/email), scheduled_at, sent_at, status.

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5. User Roles & Permissions

The system defines four user roles, each with a specific set of permissions.

Permission	Owner/Admin	Receptionist	Staff Member	Customer
View all appointments	✓	✓	Own only	Own only
Create appointment	✓	✓	✗	✓ (online)
Edit any appointment	✓	✓	✗	✗
Cancel appointment	✓	✓	✗	✓ (own)
View customer profiles	✓	✓	Own clients	Own profile
Edit customer profiles	✓	✓	✗	Own profile
View payments	✓	✓	✗	Own only
Record/edit payments	✓	✓	✗	✗
Manage employees	✓	✗	✗	✗
Manage services	✓	✗	✗	✗
View reports	✓	Limited	✗	✗
Approve bookings	✓	✓	✗	✗
Delete records	✓	✗	✗	✗

This proposal was prepared based on a domain-discovery interview with the client. All domain models, database structures, and permission matrices are subject to revision following client review. No code or technical implementation is included in this document.